

SIC Smart Bank System

SIC 8
CP Course
Team 7

Agenda

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- | Problem Statement

- | Methodology

- | Key Tasks

- | Results and Insights

- | Conclusion

- | Q&A

| Team

- Hassan Badr
- Ahmed Tarek
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| Introduction

- Project Overview: A console-based banking application (**SIC SMART BANK SYSTEM**) implemented in Python using file-based storage.
- Core Objective: Provide safe account registration, role-based login (**Admin vs. Standard User**), financial operations, and analytics tools.
- Key Technologies: Persistent **JSON storage**, 2D matrix manipulation, deep object copying, and Python set operations.

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| Problem Statement

- **Insecure Account Management:** Need for structured input validation during sign-up to eliminate duplicate phone numbers and emails.
- **Data Persistence Risks:** Requiring immediate and consistent sync between local memory operations and file storage (users.json).
- **Role Isolation:** Ensuring standard users cannot access sensitive administrative features like user segment analytics and ATM matrix controls.

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| Methodology

- **Data Layer:** Utilizes users.json for persistence, leveraging json.load() at startup and a dedicated save_users() helper for json.dump() calls.
- **Authentication Flow:** Implements a lookup loop for integer ID checking followed by a 3-attempt password verification loop.
- **Role-Based Control:** Branching application flow based on `curent_user["profile"]["stat"]` to serve either Admin or User interaction menus.

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| Key Tasks

- **Account Registration & Authentication:** Validates non-empty fields, checks duplicate phone/email entries, generates auto-incrementing IDs, and enforces login attempt limits.
- **Banking Operations:** Handles currency conversions (SAR x 9, USD x 30 to EGP), validates withdraw balances, and performs dual-account transfers with transaction logging.
- **Admin Analytics & ATM Management:** Implements set theory (union, intersection, difference) for reporting and manages a 2D matrix to track machine availability.

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| Results and Insights

- **Multi-Currency Processing:** Deposit conversion automatically updates local wallet balances while preserving detailed audit trail messages in history.
- **Set-Based Duplicate & Segment Analysis:** Efficiently detects duplicate user contact details and generates audience intersections (e.g., Active vs. VIP users).
- **State Preservation during Profile Edits:** Uses `copy.deepcopy()` to create a historical profile snapshot prior to applying `dict.update()` or `dict.pop()` modifications.

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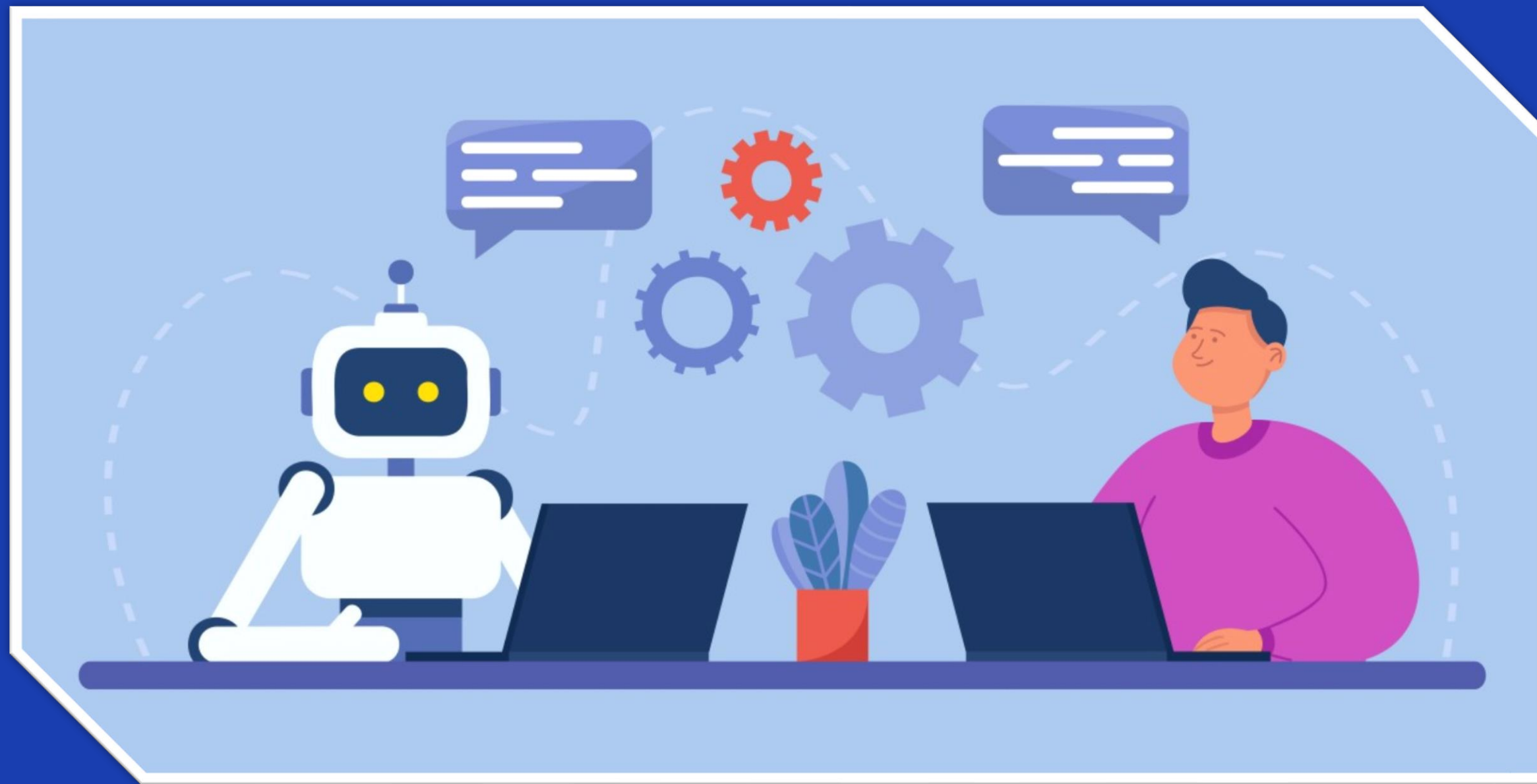
| Conclusion

- **Functional Success:** Delivered a feature-complete banking prototype executing core operations, reporting, and matrix grid updates.
- **Technical Takeaway:** Demonstrates effective practical usage of native Python data structures (lists, dicts, sets, 2D arrays) and JSON file I/O.
- **Future Work:** Recommendations include transitioning to password hashing, implementing database concurrency (SQLite/SQL), and separating monolithic code into modular classes.

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Do You Have any Questions?



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